

ROSHAN SHARMA

| Aspiring Robotics & Cloud/DevOps Engineer |

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[My Portfolio](#)

CAREER OBJECTIVE

Final-year B.Tech. Robotics Engineering student at Garden City University with hands-on experience in autonomous systems, ROS2, and computer vision. Actively developing skills in cloud computing (AWS), DevOps (Docker, Kubernetes, Terraform), and CI/CD pipelines. Published research on AI-based vehicle detection using YOLOv5 and Deep SORT. Seeking roles in Robotics Engineering or Cloud/DevOps where I can contribute to scalable, intelligent systems.

EDUCATION

Garden City University, Bengaluru <i>BTech. in Robotics Engineering 7th Semester CGPA: 8.7 / 10.0</i>	2023 – 2027
Indiranagar Composite PU College, Bengaluru <i>12th Standard – PCMB (Science) 83%</i>	2021 – 2023
HAL East Primary and High School, Bengaluru <i>10th Standard – SSLC 67%</i>	Completed 2021

TECHNICAL SKILLS

Programming:	Python, C++, JavaScript (Learning)
Robotics:	ROS2, Gazebo Simulation, Embedded Systems, OpenCV, YOLOv5
Databases:	MySQL, MongoDB
Tools & OS:	Linux, Git, GitHub, VS Code, GitHub Actions
Cloud & DevOps:	AWS (EC2, S3, Lambda), Docker, Kubernetes, Terraform, Jenkins, CI/CD
Soft Skills:	Team Building, Communication, Creativity, Adaptability, Problem Solving

RESEARCH PUBLICATION

AI-Based Vehicle Detection, Tracking & Behaviour Analysis <i>Co-authored with Shreyas N.R. (23BTRE142) Garden City University Submitted for Publication</i>	Sep 2025
- Developed a real-time traffic monitoring system combining YOLOv5s for object detection and Deep SORT for multi-vehicle tracking. - Implemented pixel-per-meter calibration for speed estimation and acceleration-based behavioral classification (accelerating/braking/normal). - System achieved 97% detection accuracy, 93% precision, and 89% recall; tested on urban CCTV footage with annotated video output. - Turnitin plagiarism check: 0% AI-generated content detected (Submission ID: trn:oid:::13000:113983196).	

PROJECTS

Quadruped Spider Robot	<i>Autonomous Navigation Garden City University</i>
- Built a 4-legged spider robot for smart indoor navigation and real-time waypoint tracking using 12 servo motors. - Implemented autonomous path planning and obstacle avoidance using sensor fusion with ultrasonic sensors. - Developed motion control algorithms in Python and C++ integrated with ROS2 framework.	

University Robot

Computer Vision | Garden City University

- Designed a vision-guided robotic model with YOLOv5-based color detection, picking logic, and motion control.
- Implemented real-time object recognition pipeline using OpenCV and Python for precise object identification.
- Integrated servo control systems enabling accurate pick-and-place operations with sub-centimeter precision.

CloudBot Command Center

Cloud Robotics & DevOps

- Engineered a containerized decentralized robotics orchestrator using ROS 2 (Jazzy) and FastAPI, integrating autonomous BFS pathfinding and real-time obstacle avoidance physics.
- Automated a full CI/CD pipeline using GitHub Actions to securely build Docker images and deploy a live, public-facing web gateway to Render PaaS.
- Implemented stateful hardware simulation for battery degradation and streamed live telemetry data to a Google Firebase Firestore NoSQL database.
- Designed a responsive, mobile-friendly command dashboard with dynamic UI state rendering and event-driven toast notifications based on real-time backend data streams.

WORK EXPERIENCE

Customer Outreach Executive

Bharath Finnovation | 2023

KreditBee – Financial Products & Services

- Conducted outbound calls to prospective customers to promote KreditBee's financial products and services.
- Maintained persuasive communication skills to consistently achieve daily conversion targets.

Event Management Coordinator

Freelance | 2024 – 2025

Brands: Swiggy Mall, Guvi|HCL, Instamart, Pincode by phonepe, Narayana Health Care

- Planned and executed on-ground promotional campaigns for multiple high-profile consumer brands.
- Coordinated teams and managed customer engagement drives, consistently achieving brand visibility targets.

CERTIFICATIONS

Courses / Certificates	Issuing Body	Date	Verify
IBM Deep Learning with PyTorch, Keras and Tensorflow	IBM · Coursera	April 2026	Coursera Certificate – View Certificate 1
Interfacing with the Raspberry Pi	UC Irvine · Coursera	March 2026	Coursera Certificate – View Certificate 2
Modern Robotics, Course 2: Robot Kinematics	Northwestern university · Coursera	March 2026	Coursera Certificate – View Certificate 3
Augmented Reality for JavaScript Developers	Infosys Springboard	February 2026	Certification – View Certificate 4
Agile Project Management	Google · Coursera	February 2025	Coursera Certificate – View Certificate 5
Artificial Intelligence and Machine Learning Training Course	Infosys Springboard	August 2025	Certification – View Certificate 6

LANGUAGES

English (Fluent) | Hindi (Native) | Kannada (Conversational)

INTERESTS

Robotics & Autonomous Systems | Cloud Computing | Travel | Art | Learning Emerging Technologies